

# Multimodal Agent for Question Answering on Slide Presentations

Columbia E6998-013 Final Project Report

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## Abstract

Slide presentations are a widely used medium for communicating complex information, yet they remain difficult to query in a grounded and multimodal way. We study an agentic approach to question answering over slide decks, where a large language model orchestrates retrieval, reranking, and perception tools to adaptively balance answer quality against latency and computational cost. Our system combines caption-based dense retrieval, multimodal late-interaction retrieval with ColPali, and targeted perception via OCR and vision language models, and is engineered to run end-to-end on a single 24GB consumer GPU. Evaluated on the SlideVQA benchmark in an open-domain setting, the agent consistently outperforms fixed retrieval pipelines at comparable average latency, while avoiding the high cost of always reranking. Ablation studies show that adaptive tool access is critical for strong performance, and qualitative execution traces reveal how the agent decomposes multi-hop queries, escalates to visual reasoning when necessary, and exposes characteristic failure modes. Overall, our results demonstrate that agentic orchestration provides a practical and effective framework for multimodal slide question answering under realistic resource constraints.

## 1 Introduction

Slide presentations are a major medium for communicating knowledge in academia and industry, but they remain difficult to search and query in a grounded way. A slide QA system must not only retrieve relevant content, but do so over mixed modalities (text, figures, charts, tables) and often combine evidence across multiple slides to answer a single question. In practice, this introduces a tight tradeoff between retrieval quality and latency or compute cost, since stronger tools (for example visual retrievers or rerankers) are slower and more expensive to run.

Most deployed solutions today rely on proprietary hosted LLMs paired with managed vector databases for multimodal embeddings, which limits transparency and control, and raises cost and privacy concerns. Prior research on slide QA and slide retrieval has largely focused on fixed pipelines [1] [2] [3], with the retrieval component studied much more deeply than end-to-end systems that jointly retrieve, reason, and generate grounded answers.

In this work, we study an agentic approach, where an LLM orchestrator answers questions by adaptively invoking tools and synthesizing evidence from retrieved slides. We build both (i) a caption-based textual index, where each slide image is captioned offline using a VLM and embedded with a strong text embedder into a Faiss index, and (ii) a visual late-interaction index using ColPali [4] and a multi-vector storage engine, along with a third variant that applies late-interaction text embeddings to slide captions. The agent is implemented as a LangGraph ReAct tool-calling system (via Ollama) with retrieval, reranking, OCR, and VLM-query tools, designed so the orchestrator remains resident in GPU memory while secondary models are swapped in only when needed, enabling deployment on a single 24GB consumer GPU.

We organize our study around three questions: whether a tool-using agent can outperform the best fixed retrieval pipeline at comparable latency and cost, how tool choice and orchestration shape the quality-latency-cost tradeoff, and whether a single 24GB GPU is sufficient for an agentic multimodal slide QA system with usable response times. Accordingly, we hypothesize that adaptive tool orchestration is the key to the quality-latency-cost tradeoff: an agent that starts with fast, cheap retrieval (captions or embeddings) and escalates only when necessary to more expensive perception (OCR or VLM-based inspection) will achieve better grounded answer quality and evidence coverage than any single fixed retrieval pipeline at comparable average latency, while remaining feasible to run end-to-end on a single 24GB consumer GPU.

## 2 Related Work

Recent progress in vision language models (VLMs) has made it practical to process slides as multimodal documents, rather than relying on brittle modality extraction. Prior work traces this trajectory from contrastive pretraining (CLIP) [5] and vision language pretraining frameworks (BLIP) [6] to strong general VLMs (for example GPT-5 and Gemini-3), and highlights a parallel trend toward smaller, more deployable models such as Gemma3 [7] and Qwen3-VL [8]. Importantly for slide understanding, newer open VLMs are trained on mixtures that include OCR, captioning, and chart or table understanding [9], enabling detailed caption generation in zero-shot settings.

On the retrieval side, [3] surveys both direct multimodal retrieval and caption-based alternatives. It discusses Document Screenshot Embeddings (DSE) [1], which encodes document screenshots directly via a VLM bi-encoder to capture layout, visual cues, and text without explicit OCR, trained with contrastive learning. It also covers ColPali [4], which adapts ColBERT-style late interaction to the multimodal setting by comparing query token embeddings against patch-level image embeddings, improving expressiveness over single-vector encoders. Finally, it motivates caption-first pipelines by connecting to evidence that retrieval over image captions can improve downstream RAG, and argues that doing so directly over whole slides avoids the need for pre-extracting and separately indexing slide elements.

## 3 Approach

We propose an agentic framework that dynamically orchestrates retrieval and reasoning over slide decks. Unlike fixed RAG pipelines, which execute a predetermined sequence of retrieval and generation steps, our system employs an LLM orchestrator that autonomously selects from a suite of specialized tools, ranging from fast textual search to expensive visual analysis, to answer user queries.

### 3.1 Offline Preprocessing and Indexing

To support multimodal retrieval, we implement three distinct indexing strategies. This preprocessing is performed offline, before the agent is deployed.

#### 3.1.1 Caption-Based Dense Indexing

We first convert the multimodal slide retrieval problem into a text retrieval task. For every slide in the corpus, we generate a detailed textual caption using a VLM. We utilize Gemma-3-27B-QAT<sup>1</sup> [7] (via Ollama) for the captioning, a quantized VLM capable of describing visual elements such as charts and layouts. These captions are then encoded into dense vectors using the state-of-the-art text embedding model NV-Embed-v2<sup>2</sup> [10] (8B parameters) (via Hugging Face). The resulting embeddings are stored in a Faiss<sup>3</sup> [11] index for efficient similarity search. This approach allows the agent to perform fast, low-cost retrieval by matching query semantics against slide captions.

<sup>1</sup><https://ollama.com/library/gemma3:27b>

<sup>2</sup><https://huggingface.co/nvidia/NV-Embed-v2>

<sup>3</sup><https://github.com/facebookresearch/faiss>

### 3.1.2 Visual Late-Interaction Indexing

To capture visual nuances lost in captioning, we employ `vidore/colpali`<sup>4</sup> [4], a ColBERT-style [12] late-interaction model adapted for multimodal data. ColPali encodes slide images into a set of patch-level visual embeddings and textual queries into token-level embeddings. Relevance is computed via MaxSim operations between query terms and image patches. We utilize Plaid<sup>5</sup> [13], a storage engine optimized for multi-vector search, to index these visual representations.

### 3.1.3 Textual Late-Interaction Indexing

As a middle ground between standard dense retrieval and visual late interaction, we implement a third index that applies the ColPali text encoder directly to the generated slide captions. This method ignores the visual encoder, leveraging the scaled-up ColBERT-style architecture of ColPali to perform late-interaction retrieval solely on the textual content of the captions. These embeddings are also stored in a Plaid index.

## 3.2 Agent Architecture

The core of our system is a LangGraph ReAct agent powered by an LLM orchestrator running via Ollama. We evaluate two open-weights models as orchestrators, `gpt-oss:20b`<sup>6</sup> [14] and `qwen3-vl:30b-a3b-instruct`<sup>7</sup> [8]. The agent receives the user query and iteratively executes a loop of thought, tool selection, and observation until it synthesizes a final answer or reaches a maximum step limit.

A critical design constraint of our work is feasibility on consumer-grade hardware. We engineer the system to run on a single NVIDIA RTX 3090 (24GB VRAM). To achieve this, the orchestrator remains resident in GPU memory, while secondary "expert" models (such as the reranker or VLM) are dynamically loaded and unloaded from VRAM only when their corresponding tools are invoked.

## 3.3 Tool Definitions

The agent has access to a diverse toolkit allowing it to balance the tradeoff between retrieval cost and answer quality.

**Retrieval Tools** The agent can invoke three primary retrieval functions corresponding to the indices described above: `retrieve_nvembed`, `retrieve_colpali_text`, and `retrieve_colpali_visual`. Each tool accepts a `top_k` parameter, returning a list of slides with their associated captions and similarity scores.

We also note that when the agent decides to call a retrieval tool, the tool encodes the question in the matching space (Faiss or Plaid index) and queries the corresponding index.

**Reranking Tool** To improve retrieval accuracy, we create a `retrieve_and_rerank` tool that performs a two-stage process. It first fetches 100 candidate slides using a specified base retriever that were described, and then scores them using the `Qwen3-Reranker-4B` [15] reranker. This model is loaded in 8-bit quantization to fit alongside the orchestrator. While computationally expensive (20s latency), this tool provides the highest retrieval quality by cross-encoding the query and slide content.

**Query Reformulation** The `custom_query_search` tool allows the agent to reformulate the search query rather than relying on the original user input. This is particularly useful for multi-hop reasoning, where the agent may need to decompose a complex question into simpler, targeted searches.

**Perception Tools** When retrieved captions are insufficient, the agent can inspect the raw slide content using two perception tools. `ocr_slide` extracts raw text from a specific slide using Tesseract OCR. This is low-cost and ideal for extracting precise numerical data or table rows not captured in the

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<sup>4</sup><https://huggingface.co/vidore/colpali>

<sup>5</sup><https://github.com/lightonai/fast-plaid>

<sup>6</sup><https://ollama.com/library/gpt-oss:20b>

<sup>7</sup><https://ollama.com/library/qwen3-vl:30b-a3b-instruct>

caption. `ask_vlm` queries a specific slide image using a VLM. If the orchestrator is multimodal (e.g., Qwen3-VL), it processes the image directly. Otherwise, a VLM (we use `moondream3`<sup>8</sup>) is swapped into memory to answer the visual query. This tool is the most expensive but essential for interpreting complex charts and visual diagrams.

Finally, a `list_tools` function enables the agent to inspect its available tools at runtime, facilitating a dynamic understanding of its environment.

## 4 Experiments

### 4.1 Dataset

We evaluate our system on the SlideVQA [2] dataset, a large-scale benchmark designed for visual question answering on slide presentations. The dataset comprises 2,619 slide decks collected from SlideShare, spanning both business and academic domains, with a total of over 52,000 individual slides. It contains approximately 14,500 question-answer pairs, where each sample consists of a natural language query, a gold standard answer, and a set of annotated "evidence" slides containing the information required to derive the answer.

The dataset presents distinct challenges suitable for agentic evaluation. It requires multimodal understanding, as answers are often embedded in charts, figures, or complex layouts rather than plain text. Furthermore, the dataset tests multi-hop reasoning capabilities; many questions require synthesizing information across multiple evidence slides or performing arithmetic operations to reach the final answer.

As done in other work that employs this dataset, we adopt a more challenging and realistic ‘open-domain’ setting where all 2,619 decks are treated as a single shared retrieval pool. This simulates a real-world repository where an agent must locate relevant information without prior knowledge of the specific source deck.

To ensure our agentic experiments remain computationally feasible on a single GPU, we evaluate performance on a sampled subset of 50 queries. However, to preserve the difficulty of the retrieval task, we build our vector indices (both caption-based and visual) using the full corpus of 52,000 slides. This ensures that the agent is tested in a "needle-in-a-haystack" scenario, where it must successfully discriminate relevant evidence from tens of thousands of distractors. Figure 1 illustrates a representative QA sample, showing a query, its corresponding answer, and the multimodal evidence slide required to answer it.

### 4.2 Evaluation method

To assess the effectiveness of our agentic approach, we employ three primary metrics covering answer quality, retrieval completeness, and system efficiency.

**Accuracy** We define accuracy as the fraction of questions for which the generated answer semantically matches the gold standard answer. Since open-ended generation can vary in phrasing, exact string matching is insufficient. Instead, we employ an LLM-as-a-Judge approach. We use Gemma3-27B-IT-QAT (4-bit quantized) as the evaluator. The judge is prompted with the question, the gold answer, and the agent’s predicted answer, and is tasked with outputting a binary label indicating whether the prediction is correct.

**Evidence Recall** To evaluate the grounding of the answers, we measure Evidence Recall. This is defined as the fraction of queries where all annotated gold evidence slides are successfully retrieved by the agent during its execution. This metric ensures the agent has located the complete context required for reasoning, rather than partial information.

**Efficiency** We report the Average Inference Time, measured as the wall-clock time from receiving the user query to generating the final response, averaged over the test set. This metric captures the latency cost of the agent’s adaptive tool usage and thinking process.

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<sup>8</sup><https://huggingface.co/moondream/moondream3-preview>

| 2011: Strong Full Year Performance                                  |             |          |                            |  |
|---------------------------------------------------------------------|-------------|----------|----------------------------|--|
|                                                                     | As reported |          | At constant exchange rates |  |
| Sales                                                               | 83.6 bn     |          |                            |  |
| Organic Growth                                                      | +7.5%       |          |                            |  |
| Real Internal Growth                                                | +3.9%       |          |                            |  |
| Trading Operating Profit                                            | 12.5 bn     |          |                            |  |
| Trading Operating Profit Margin<br>(vs. 2010 continuing operations) | 15.0%       | +60 bps  | +90 bps                    |  |
| Net Profit<br>(vs. 2010 continuing operations)                      | 9.5 bn      | +8.1%    |                            |  |
| Net Profit Margin<br>(vs. 2010 continuing operations)               | 11.3%       | +130 bps |                            |  |
| Underlying EPS<br>(vs. 2010 Group)                                  | 3.08        |          | +7.8%                      |  |
| Dividend<br>(proposed)                                              | 1.95        | +5.4%    |                            |  |

All figures in CHF

4 | February 21st, 2012 | 2011 Full Year Results Roadshow

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Figure 1: A representative example from the SlideVQA dataset. The query is ‘How much is the Trading Operating Profit in 2011?’ and the answer is ‘12.5 bn’.

### 4.3 Experimental details

We first compare our agent against non-agentic baselines that use fixed retrieval pipelines. These baselines employ the same underlying retrieval models (NV-Embed, ColPali-Text, ColPali-Visual) with and without reranking, directly feeding the top-10 results to the LLM for generation (in case of a reranker, 100 samples are initially retrieved, and are then reduced to the top-10). To establish performance upper bounds, we also evaluate an Oracle baseline where the gold evidence slides are provided directly to the model.

We conduct a comprehensive study on the agentic framework by systematically disabling individual tools from the full configuration to isolate their specific contributions to accuracy and latency. We also evaluate a "Minimal" configuration that restricts the agent to only the three basic retrieval tools, removing access to all advanced capabilities. This breakdown allows us to analyze the trade-offs between tool complexity and performance, specifically identifying which components drive evidence recall and answer accuracy versus those that incur disproportionate computational costs.

### 4.4 Results

We present our experimental results in two parts: first, a comparison of non-agentic baselines using fixed retrieval pipelines, and second, a detailed ablation study of the agentic framework.

#### 4.4.1 Non-Agentic Baselines

Table 1 summarizes the performance of fixed retrieval pipelines. We observe a stark trade-off between latency and accuracy. Standard retrieval methods (NV-Embed and ColPali) are highly efficient, with inference times ranging from 1.6s to 3.7s, but their accuracy caps at 27/50. Introducing a reranker often improves accuracy (e.g., improving gpt-oss:20b with NV-Embed from 25/50 to 30/50), but this comes at a significant computational cost, increasing latency to  $\sim 30$  seconds per query. The Oracle baseline demonstrates the theoretical upper bound of our evaluator models, achieving 33/50 and 35/50 accuracy for GPT and Qwen, respectively.

Table 1: Performance of non-agentic fixed retrieval pipelines.

| Model               | Retrieval Method      | Accuracy     | Recall       | Time (s)   |
|---------------------|-----------------------|--------------|--------------|------------|
| gpt-oss:20b         | NV-Embed              | 25/50        | 34/50        | 3.4        |
| gpt-oss:20b         | ColPali-Text          | 25/50        | 41/50        | 3.7        |
| gpt-oss:20b         | ColPali-Visual        | 27/50        | 44/50        | 3.5        |
| gpt-oss:20b         | NV-Embed + Rerank     | 30/50        | 43/50        | 30.1       |
| gpt-oss:20b         | ColPali-Text + Rerank | 27/50        | 42/50        | 29.8       |
| <i>gpt-oss:20b</i>  | <i>Oracle (Gold)</i>  | <b>33/50</b> | <b>50/50</b> | <b>1.6</b> |
| qwen3-vl:30b        | NV-Embed              | 22/50        | 34/50        | 1.6        |
| qwen3-vl:30b        | ColPali-Text          | 24/50        | 41/50        | 1.6        |
| qwen3-vl:30b        | ColPali-Visual        | 26/50        | 44/50        | 1.6        |
| qwen3-vl:30b        | NV-Embed + Rerank     | 25/50        | 43/50        | 28.6       |
| qwen3-vl:30b        | ColPali-Text + Rerank | 23/50        | 42/50        | 28.3       |
| <i>qwen3-vl:30b</i> | <i>Oracle (Gold)</i>  | <b>35/50</b> | <b>50/50</b> | <b>0.6</b> |

#### 4.4.2 Agentic Results and Ablation

Table 2 presents the results of the agentic system. The "All Tools" configuration achieves competitive accuracy (29/50 for GPT and 32/50 for Qwen) with moderate latency ( $\sim 16$ s), effectively balancing the speed of dense retrieval with the precision of heavier tools.

Table 2: Agentic ablation study. We report Accuracy, Evidence Recall, Average Inference Time, and Average Tool Calls per query.

| Model        | Experiment       | Acc.         | Recall       | Time (s)    | Calls      |
|--------------|------------------|--------------|--------------|-------------|------------|
| gpt-oss:20b  | All Tools        | 29/50        | 41/50        | 16.2        | 4.3        |
| gpt-oss:20b  | w/o Reranker     | 25/50        | 41/50        | 14.0        | 4.0        |
| gpt-oss:20b  | w/o VLM          | 25/50        | <b>42/50</b> | 31.8        | 4.6        |
| gpt-oss:20b  | w/o Custom Query | 29/50        | 38/50        | <b>11.4</b> | <b>3.3</b> |
| gpt-oss:20b  | w/o OCR          | 27/50        | 42/50        | 14.6        | 3.8        |
| gpt-oss:20b  | w/o ColPali      | <b>30/50</b> | 41/50        | 15.2        | 4.0        |
| gpt-oss:20b  | w/o NV-Embed     | 27/50        | 41/50        | 28.8        | 3.7        |
| gpt-oss:20b  | Minimal          | 21/50        | 36/50        | 17.9        | 4.2        |
| qwen3-vl:30b | All Tools        | 32/50        | 36/50        | 15.8        | 4.9        |
| qwen3-vl:30b | w/o Reranker     | <b>34/50</b> | 38/50        | 14.0        | 4.3        |
| qwen3-vl:30b | w/o VLM          | 26/50        | <b>40/50</b> | 14.4        | 4.7        |
| qwen3-vl:30b | w/o Custom Query | 31/50        | 39/50        | 35.3        | 4.4        |
| qwen3-vl:30b | w/o OCR          | 34/50        | 38/50        | 12.3        | 3.9        |
| qwen3-vl:30b | w/o ColPali      | 33/50        | 40/50        | 17.4        | 4.7        |
| qwen3-vl:30b | w/o NV-Embed     | 33/50        | 38/50        | 14.2        | 4.4        |
| qwen3-vl:30b | Minimal          | 19/50        | 37/50        | <b>7.2</b>  | <b>3.1</b> |

**Impact of Tools** Our ablation study confirms that tool access is critical. The "Minimal" configuration, which restricts the agent to basic retrieval, results in a significant performance drop, reducing

accuracy from 30/50  $\rightarrow$  21/50 for GPT and 34/50  $\rightarrow$  19/50 for Qwen. This confirms that adaptive access to advanced tools is essential for handling complex queries.

### Specific Component Analysis

- **Custom Query:** Excluding the "Custom Query" tool generally saves time without significantly hurting accuracy. For `gpt-oss:20b`, excluding this tool results in the lowest average time (11.4s) and average tool calls (3.3), suggesting the model can often answer efficiently without reformulating the search.
- **OCR:** Removing the OCR tool has a negligible impact on performance ( $-2$  accuracy for GPT,  $+2$  for Qwen), indicating that our VLM-generated captions already capture the necessary textual information from the slides.
- **VLM and Recall:** Interestingly, Evidence Recall and Accuracy do not always correlate. For both models, the highest Evidence Recall is achieved by *excluding* the "VLM Query" tool (42/50 for GPT, 40/50 for Qwen). This suggests that while visual inspection tools help with specific reasoning, they may occasionally cause the agent to terminate its search prematurely, limiting the breadth of retrieved evidence.

**Model-Specific Findings** The two orchestrators exhibit different optimal configurations. For `gpt-oss:20b`, the best accuracy (30/50) is achieved by excluding ColPali queries, potentially because the visual retrieval introduces distractors that confuse the smaller model. Conversely, for `qwen3-v1:30b`, the best accuracy (34/50) is achieved by excluding the Reranker, while the lowest latency is achieved in the Minimal configuration.

## 5 Analysis

The traces in Appendix Tables 3 through 6 provide a qualitative view of how agentic orchestration actually plays out beyond aggregate Accuracy and Evidence Recall. Across these examples, we see a consistent pattern that matches the intended design. Agents begin with cheap retrieval and only escalate to heavier tools when the retrieved representation (typically captions) is insufficient, reflecting the core quality vs latency tradeoff described in our methodology.

First, Tables 3 and 4 illustrate two successful behaviors. In the simple case (Table 3), the agent retrieves the golden slide and answers immediately from the caption, demonstrating that "fast path" retrieval can be enough when captions are faithful. In the multi-hop case (Table 4), the agent decomposes the question into an intermediate subgoal (identify the acquisition year), verifies it via targeted retrieval and OCR, and then returns to the performance slide to complete the comparison. This trace highlights how tool calls are not just retrieval improvements but also a mechanism for temporal grounding and structured subproblem solving.

The failure and "skeptical success" cases in Tables 5 and 6 expose complementary failure modes and strengths. Table 5 shows a classic agentic pathology. Despite already having sufficient evidence from retrieval and OCR, the agent over-weights an unreliable VLM response ("No answer"), repeatedly re-invokes the same tool, and fails to synthesize a final answer, despite having all the relevant context. In contrast, Table 6 shows the opposite and more desirable behavior. The agent recognizes that both the caption and OCR are inadequate for reading chart values, refuses to guess, and escalates to visual inspection to extract the needed numbers. Together, these two traces motivate future work on (i) loop prevention and termination criteria, and (ii) uncertainty-aware tool arbitration so that agents can trust strong evidence when it exists, and escalate only when it truly adds information.

## 6 Practical Impact and Implications

A key practical outcome of this work is that an end-to-end, tool-using multimodal slide QA system can be deployed on a single consumer RTX 3090 (24GB) without relying on proprietary hosted LLMs or managed multimodal vector infrastructure. In our setup, the expensive preprocessing stages (slide captioning and caption embedding) are feasible on the same 3090 used at inference time, while inference is engineered so the orchestrator remains resident in VRAM and only one "expert" model is swapped in when needed. This improves resource efficiency and makes the system realistic for

real-world settings such as enterprise slide repositories, research groups and classrooms (searching lecture decks), and compliance-sensitive organizations that cannot upload proprietary decks to third parties. From a user experience perspective, the agent’s ability to start with fast retrieval and escalate selectively supports more interactive behavior than “always rerank” pipelines, while still enabling grounded answers tied to specific retrieved slides.

More broadly, these results suggest that agentic orchestration is a viable systems strategy for multi-modal QA under tight memory and cost constraints, not just a modeling choice. The “swap-in experts” design implies a scalable path where organizations can keep the default experience low-latency (cheap retrieval and caption grounding) and reserve expensive perception or reranking for harder queries, rather than paying that cost on every request. This architecture also offers a clean extension point for future deployments: additional tools (for example domain-specific retrievers or citation validators) can be integrated without changing the core interaction loop, while maintaining the same hardware envelope by enforcing the constraint that the GPU holds the orchestrator plus at most one auxiliary model at a time.

## 7 Limitations and Future Work

A primary limitation of our system is that tool usage is not optimized or explicitly taught. The agent selects tools in an ad-hoc manner based on the base orchestrator’s prompting and general reasoning, which can lead to inefficient or even pathological behaviors (e.g., unnecessary escalations or repeated calls). Future work could explore tool-use fine-tuning or stronger few-shot prompting to explicitly teach when and how to use each tool, for example guidelines like ‘use OCR primarily for extracting precise numbers from charts and tables’ or ‘only invoke VLM inspection when caption and OCR disagree or are incomplete.’

A second limitation is that our retrieval stack explores only a narrow set of representation choices. We rely on ColPali for visual late interaction and a captioning based dense index for fast retrieval, but other image encoders could provide better robustness to chart and layout-heavy slides. Likewise, sparse retrieval over captions (e.g., BM25 style search) could serve as a complementary low-cost tool, especially for keyword-heavy queries. Finally, the current agent always produces an answer, even when evidence is weak. Future work should add uncertainty awareness, either through confidence estimation from retrieval and tool signals or by enabling explicit abstention (e.g., “I don’t know”) when the retrieved evidence does not support a grounded response.

## 8 Conclusion

We presented an open-weights, single-GPU agentic system for question answering over slide presentations that combines multiple retrieval indices (caption-based dense search and ColPali late interaction) with optional reranking and targeted perception tools (OCR and VLM-based inspection). By keeping the orchestrator resident in VRAM and swapping in expert models only when needed, our design demonstrates that end-to-end multimodal slide QA with adaptive tool use is feasible on consumer hardware.

Experiments on a challenging open-domain SlideVQA setting show a clear quality–latency tradeoff for fixed pipelines. Fast retrievers are efficient but accuracy-limited, while reranking improves accuracy at substantial latency cost. In contrast, the agent can balance this tradeoff by selectively escalating to heavier tools, achieving competitive accuracy with moderate latency, and our ablations confirm that removing certain tool access significantly degrades performance. Finally, our qualitative traces highlight both the strengths of orchestration (multi-hop decomposition and selective perception) and key failure modes (repetitive tool loops and miscalibrated trust), motivating future work on better tool-use training, stronger retrieval representations, and uncertainty-aware answering.

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## A Appendix

Table 3: Agent Execution Trace for a query. The query is ‘How much is the Trading Operating Profit in 2011?’. The agent is gpt-oss:20b, and it successfully retrieved and answered the correct answer, which is 12.5bn

| Step | Tool / Action       | Input / Arguments                        | Observation / Reasoning                                                                                                                                                                                                                  |
|------|---------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | retrieve_and_rerank | <i>Method:</i> neural<br><i>Top_k:</i> 5 | <b>Result:</b> Retrieves 5 slides. <b>Slide 1 (Golden)</b> is titled "2011: Strong Full Year Performance". The caption explicitly lists the financial breakdown, including the line item: "Trading Operating Profit: 12.5 bn".           |
| 2    | <b>Final Answer</b> | <b>12.5 bn</b>                           | <i>Conclusion:</i> The agent identifies the exact value "12.5 bn" directly from the caption of the first retrieved slide. No further perception tools (like OCR or VLM) were required as the caption contained the necessary data point. |

Table 4: Agent Execution Trace for a query. The query is ‘Which achieved better performance in the year Nestlé acquired Prometheus, Organic Growth or Real Internal Growth?’. The agent is gpt-oss:20b, and it successfully identifies the acquisition year (2011) and compares the financial metrics from the retrieved evidence.

| Step | Tool / Action       | Input / Arguments                                             | Observation / Reasoning                                                                                                                                                                                                  |
|------|---------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | retrieve_and_rerank | <i>Query:</i> Original user question                          | <b>Result:</b> Retrieves 5 slides. <b>Slide 5 (Golden)</b> contains a table for "2011: Strong Full Year Performance" showing Organic Growth (+7.5%) and Real Internal Growth (+3.9%).                                    |
| 2    | custom_query_search | <i>Query:</i> "Nestlé acquired Prometheus year"               | <b>Result:</b> Retrieves 5 slides. <b>Slide 12 (Golden)</b> shows a timeline of acquisitions from 2000–2011.                                                                                                             |
| 3    | retrieve_nvembed    | <i>Top_k:</i> 10                                              | <b>Result:</b> Retrieves broad context, including Slide 6 (Nestlé Model 2011) and Slide 9 (Shareholder returns).                                                                                                         |
| 4    | <b>Reasoning</b>    | —                                                             | <i>Thought:</i> The agent identifies the need to pinpoint the specific year Nestlé acquired Prometheus to compare the metrics. It hypothesizes the year might be 2011 based on initial retrieval but seeks confirmation. |
| 5    | custom_query_search | <i>Query:</i> "Nestlé acquired Prometheus 2011"               | <b>Result:</b> Retrieves 5 slides. Slide 12 appears again, reinforcing the timeline context.                                                                                                                             |
| 6    | ocr_slide           | <i>Deck:</i> 2012-02-20...<br><i>Slide:</i> slide_12_1024.jpg | <b>Result:</b> Extracts text from the acquisition timeline slide: "Significant acquisitions added new growth platforms... Acquisitions 2000 to 2011".                                                                    |
| 7    | custom_query_search | <i>Query:</i> "Prometheus Nestlé 2011"                        | <b>Result:</b> Retrieves 10 slides, further confirming the 2011 context with financial results (Slide 5) and acquisition history (Slide 12).                                                                             |
| 8    | <b>Final Answer</b> | <b>Organic Growth</b>                                         | <i>Conclusion:</i> Based on Slide 5 (retrieved in Step 1 and confirmed relevant by the 2011 date established in Steps 2-7), Organic Growth (7.5%) was higher than Real Internal Growth (3.9%).                           |

Table 5: Agent Execution Trace for query. The query is ‘How much is the Trading Operating Profit in 2011?’. The answer is ‘12.5bn’. The agent is qwen3-v1:30b-a3b-instruct. Despite correctly retrieving the evidence slide and extracting the answer via OCR, the agent fails to answer this simple question, getting stuck in a repetitive loop of failed VLM queries.

| Step | Tool / Action         | Input / Arguments                                                | Observation / Reasoning                                                                                                                                                                                                             |
|------|-----------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | retrieve_and_rerank   | <i>Method:</i> neural<br><i>Top_k:</i> 5                         | <b>Result:</b> Retrieves Slide 5 (Golden Evidence). The caption explicitly states: "Trading Operating Profit: 12.5 bn".                                                                                                             |
| 2    | ocr_slide             | <i>Deck:</i> ...roadshow...<br><i>Slide:</i> slide_5_1024.jpg    | <b>Result:</b> OCR successfully extracts the text "Trading Operating Profit... 12.5 bn" from the slide image.                                                                                                                       |
| 3    | ask_vlm               | <i>Question:</i> "What is the Trading Operating Profit in 2011?" | <b>Result:</b> The VLM returns "No answer". Despite having the answer from Steps 1 and 2, the agent distrusts the text tools and attempts visual verification.                                                                      |
| 4-8  | <b>Loop / Failure</b> | ask_vlm (Repeated)                                               | <i>Error:</i> The agent enters a failure loop, repeatedly invoking ask_vlm with the exact same parameters and receiving "No answer" each time. It fails to synthesize the available evidence and terminates without a final answer. |

Table 6: Agent Execution Trace for query. The query is ‘In which year did Nestlé achieve higher Organic Growth, 2003 or 2004?’. The agent is qwen3-v1:30b-a3b-instruct. Critically, the agent recognizes that the retrieved caption is wrong and that OCR fails to extract the data points from the chart. Instead of hallucinating, it correctly identifies the need for visual perception and pivots to using the VLM. For the same query, gpt-oss:20b trusts the (wrong) caption of the slide and returns the wrong number

| Step | Tool / Action    | Input / Arguments                                                                        | Observation / Reasoning                                                                                                                                                                                                                                |
|------|------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | retrieve_nvembed | <i>Top_k:</i> 5                                                                          | <b>Result:</b> Retrieves Slide 7 (Golden Evidence). The caption identifies a “Stacked Column Chart” for Organic Growth but the text description of the data series is truncated and insufficient for comparison.                                       |
| 2    | ocr_slide        | <i>Deck:</i> ...roadshow...<br><i>Slide:</i> slide_7_1024.jpg                            | <b>Result:</b> OCR extracts general labels like “10-year annual averages” and “Organic Growth 6.4%”, but fails to capture the specific bar values for 2003 and 2004.                                                                                   |
| 3    | <b>Reasoning</b> | —                                                                                        | <i>Thought:</i> The agent explicitly evaluates the tool output: “I see the OCR text... but it doesn’t contain the specific Organic Growth percentages... I need to get the actual values from the chart.” It decides to escalate to visual inspection. |
| 4    | ask_vlm          | <i>Question:</i> “What were the Organic Growth percentages for Nestlé in 2003 and 2004?” | <b>Result:</b> The agent attempts to read the chart visually and it succeeds. It then outputs the correct answer.                                                                                                                                      |